Department of Computer Science

Course: COMP-443 (Deep Learning)

ASSIGNMENT 03

Implementation and Performance Comparison of ResNet and GoogLeNet/Inception for Image Classification

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Instructor	Dr. Arshad Iqbal
Program	B.S AI - Red
Semester	6th
Dataset	CIFAR-10
Models	ResNet50 and InceptionV3 (Transfer Learning)
Framework	TensorFlow / Keras
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Transfer Learning on CIFAR-10: ResNet50 vs InceptionV3

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ABSTRACT

Abstract— This assignment implements and compares two pretrained convolutional neural network backbones, ResNet50 and InceptionV3 (GoogLeNet/Inception family), for CIFAR-10 classification using transfer learning. A two-stage protocol is used: (i) train only the new classifier head while freezing the ImageNet backbone, followed by (ii) fine-tuning by unfreezing the last N backbone layers with a smaller learning rate while keeping BatchNorm layers frozen. Results are evaluated using test accuracy/loss, training curves, confusion matrices, and per-class precision/recall/F1.

Keywords— CIFAR-10, Transfer Learning, Fine-tuning, ResNet50, InceptionV3, Confusion Matrix, Classification Report.

I. OBJECTIVE AND TASKS

The objective is to implement and compare ResNet50 and GoogLeNet/Inception (InceptionV3) using transfer learning on a standard dataset, and report differences in accuracy, loss, training time, and model size. The expected artifacts include training curves, confusion matrices, and a short report.

Notebook: `Assignment_03_Image_Classification_ResNet50_vs_InceptionV3_CIFAR10.ipynb`
Organized outputs: `figures/`, `results/`, `models/`, `reports/`.

II. DATASET AND PREPROCESSING

CIFAR-10 contains 60,000 RGB images (32×32) across 10 classes. Since ImageNet pretrained models expect larger inputs, CIFAR-10 images are resized to 224×224 for ResNet50 and 299×299 for InceptionV3. Model-specific preprocessing functions are applied to match the ImageNet training distribution.

Important Code Snippet: tf.data Pipeline

```
def make_dataset(x, y, image_size, preprocess_fn, training, batch_size, name):
    ds = tf.data.Dataset.from_tensor_slices((x, y))
    if training:
        ds = ds.shuffle(10_000, seed=SEED, reshuffle_each_iteration=True)

    def _map(img, label):
        img = tf.cast(img, tf.float32)
        if training:
            img = augment(img)
        img = tf.image.resize(img, image_size, method="bilinear")
        img = preprocess_fn(img)
        return img, tf.one_hot(label, num_classes)

    ds = ds.map(_map, num_parallel_calls=AUTOTUNE)
    ds = ds.batch(batch_size).prefetch(AUTOTUNE)
    return ds
```

III. MODEL ARCHITECTURES AND TRANSFER LEARNING SETUP

Both models use ImageNet weights with the top classification layer removed (`include_top=False`) and global average pooling. A dropout layer is used for regularization, followed by a dense softmax classifier sized to the CIFAR-10 class count (10 classes).

Important Code Snippet: Backbone + New Classification Head (ResNet50)

```
backbone = ResNet50(weights="imagenet", include_top=False, pooling="avg")
backbone.trainable = False

x = backbone(inputs, training=False)
x = layers.Dropout(0.3)(x)
outputs = layers.Dense(num_classes, activation="softmax", dtype="float32")(x)
```

IV. TRAINING STRATEGY

Training is performed in two stages for stability and to reduce overfitting: head training with a frozen backbone, then fine-tuning by unfreezing the last N layers with a lower learning rate. BatchNorm layers are kept frozen during fine-tuning to avoid instability on small batches.

TABLE I
TWO-STAGE TRAINING PROTOCOL

Stage	Trainable Parameters	Learning Rate	Epochs	Notes
Head Training	Classifier head only	1e-3	8	Backbone frozen
Fine-tuning	Last N backbone layers + head	1e-4	7	BatchNorm frozen

Important Code Snippet: Unfreeze Last N Layers (BN Frozen)

```
def set_finetune(backbone, unfreeze_last_n):
    backbone.trainable = True
    for layer in backbone.layers[:-unfreeze_last_n]:
        layer.trainable = False
    freeze_batchnorm(backbone)
```

TABLE II
KEY HYPERPARAMETERS (THIS RUN)

Setting	ResNet50	InceptionV3
Input size	224×224	299×299
Batch size	64	32
Epochs (head + fine-tune)	8 + 7	8 + 7
Unfreeze depth (last N layers)	40	60
Learning rates (head → fine-tune)	1e-3 → 1e-4	1e-3 → 1e-4

V. EVALUATION METRICS

The models are evaluated on the CIFAR-10 test set using accuracy and cross-entropy loss. To understand class-wise behavior, confusion matrices and classification reports (precision, recall, and F1-score per class) are computed.

Important Code Snippet: Confusion Matrix + Report

```
y_true, y_pred = predict_labels(model, test_ds)
cm = confusion_matrix(y_true, y_pred)
print(classification_report(y_true, y_pred, target_names=class_names))
```

VI. RESULTS AND COMPARISON

This run produced the metrics saved in `results/metrics.json` and `results/summary_table.csv`. ResNet50 achieved higher test accuracy and lower test loss compared to InceptionV3.

TABLE III
TEST-SET COMPARISON SUMMARY

Model	Test Accuracy	Test Loss	Training Time (min)	Model Size (MB)	Params	Input Size
ResNet50	0.9398	0.1936	9.98	211.45	23,608,202	224×224
InceptionV3	0.8988	0.3015	5.22	164.84	21,823,274	299×299

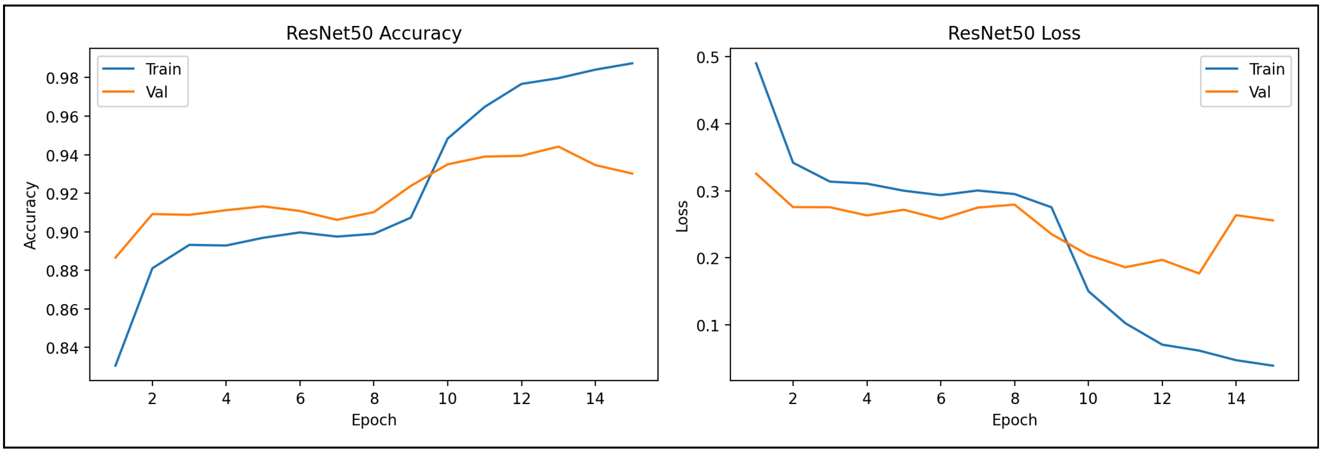

Fig. 1. ResNet50 training/validation accuracy and loss across epochs.

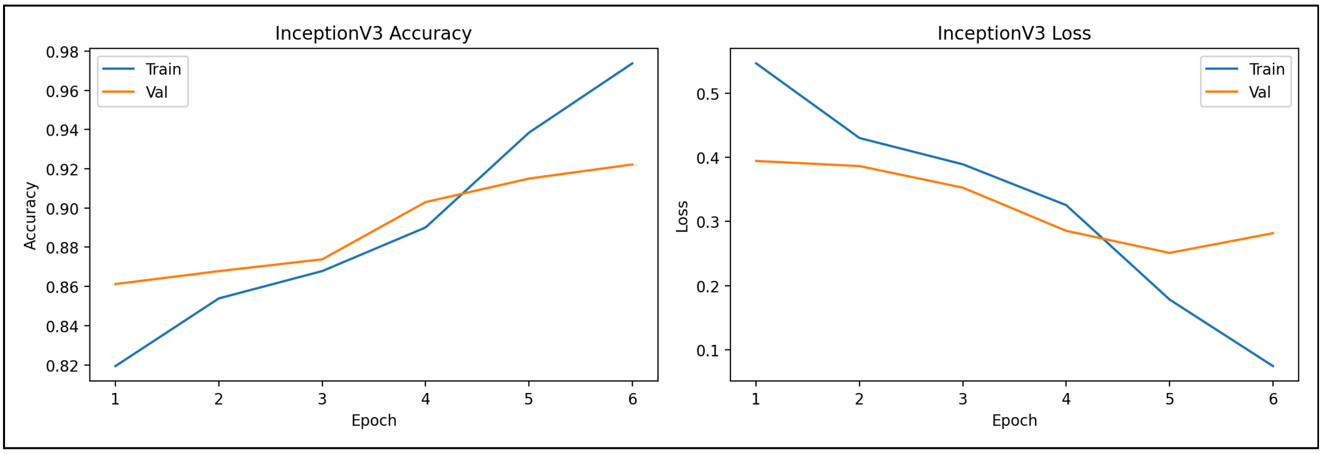

Fig. 2. InceptionV3 training/validation accuracy and loss across epochs.

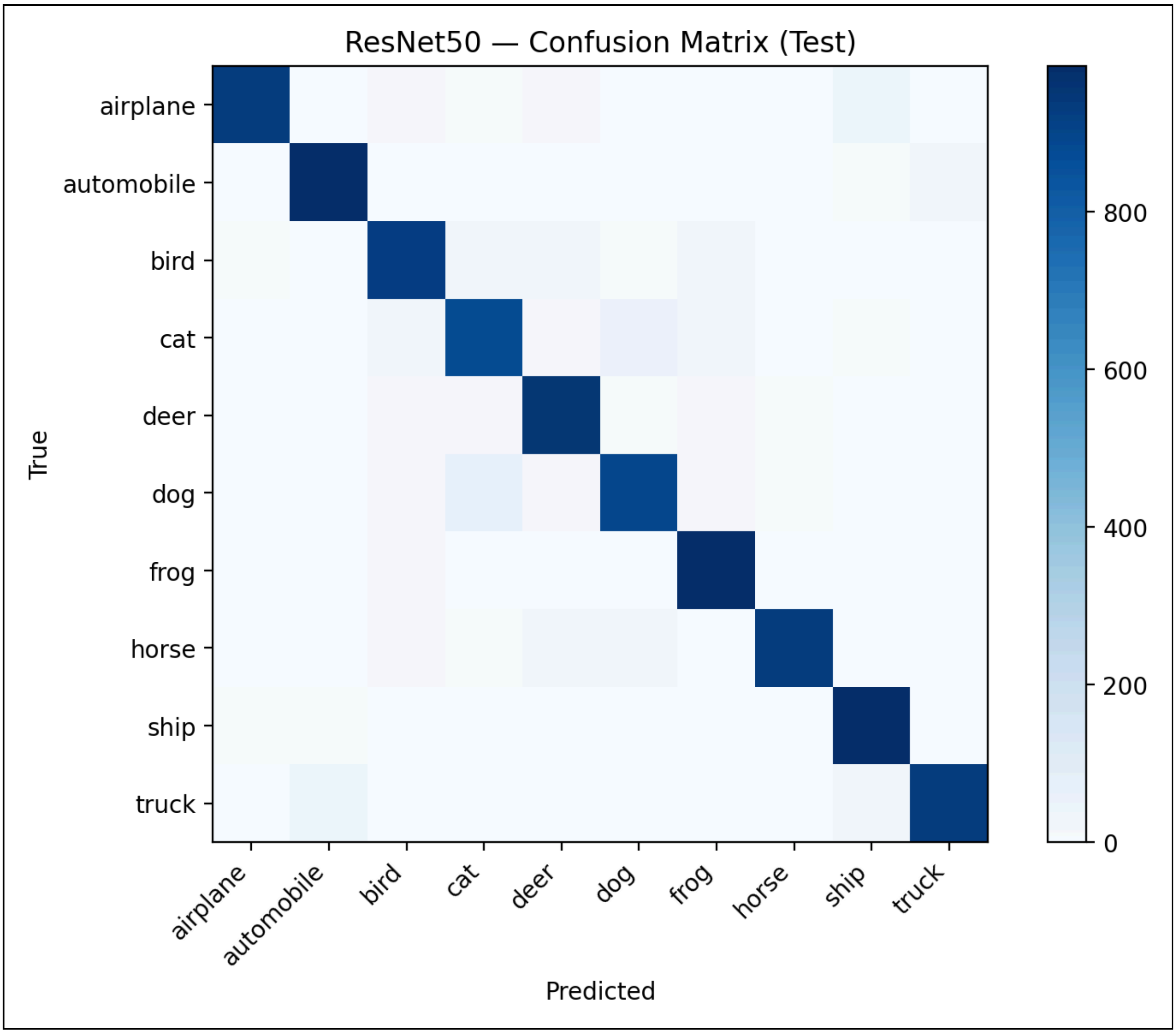

Fig. 3. ResNet50 confusion matrix on the CIFAR-10 test set.

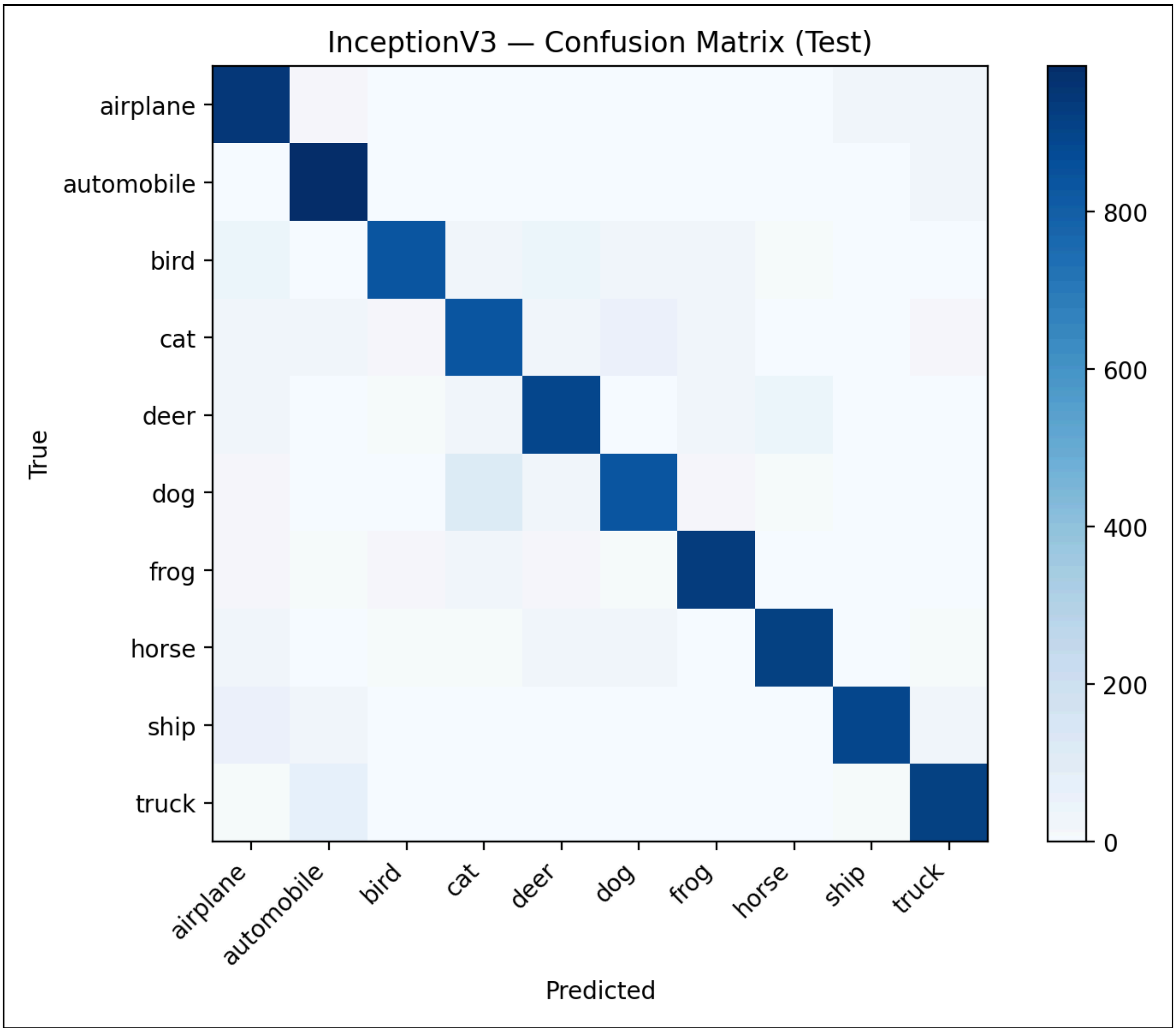

Fig. 4. InceptionV3 confusion matrix on the CIFAR-10 test set.

Per-class observations

The lowest F1-score classes indicate which categories were most challenging. For ResNet50: cat (0.8787), dog (0.9041), bird (0.9266). For InceptionV3: cat (0.8246), dog (0.8657), deer (0.8811). These patterns are consistent with CIFAR-10, where visually similar animals commonly confuse classifiers after upscaling from 32×32.

VII. DISCUSSION: WHY ONE MODEL PERFORMED BETTER

ResNet50 outperformed InceptionV3 on test accuracy and loss in this experiment. A likely reason is that the residual architecture often fine-tunes robustly under moderate augmentation and limited epochs, while InceptionV3 may be more sensitive to preprocessing, input resizing, or the selected unfreeze depth. Additionally, CIFAR-10's low resolution can reduce the benefit of the larger 299×299 input, while still increasing compute cost.

In practice, the choice depends on constraints: InceptionV3 was smaller and faster to train here, while ResNet50 produced the best test accuracy. For deployment, one must balance accuracy, latency, memory, and storage size.

VIII. SUBMISSION ORGANIZATION

Folder structure:

- `figures/`: curves and confusion matrices
- `results/`: metrics, histories, classification reports
- `models/`: saved Keras models (*.keras)
- `reports/`: archived old reports

CONCLUSION

This assignment demonstrates transfer learning with staged fine-tuning on CIFAR-10. Both pretrained architectures achieved strong performance, while ResNet50 performed best on this run. Future improvements could include stronger augmentation, label smoothing, a learning-rate schedule tuned per backbone, and systematic exploration of unfreeze depth and batch size.

REFERENCES

[1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *CVPR*, 2016.
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[3] A. Krizhevsky, “Learning Multiple Layers of Features from Tiny Images,” 2009 (CIFAR-10).